T-97701 / C4MBI67 CRITICAL CORE
Exact reconstruction of BLPTE67 and the missing T-97701 publication
Agentic Polymath Project - 18 August 2026

STATUS
The identities and equivalences in this manuscript are proved exactly. The one-sided inequality C4MBI67 is not proved. Consequently the Riemann Hypothesis remains unproved. This packet corrects a publication failure: a PR #590 comment announced T-97701 materials that were not actually present in GitHub or in the published T-97700 ZIP.

1. FROZEN SCOPE
The reconstruction starts from PR #590 at head 4f1283c67f0d4a4b504badd4c4113ba227521162. It retains the T-97700 complete small-prime cube, largest-prime Bellman identity, and terminal Type-I closure. It adds the missing critical-core theorem, exact checker, manuscript, and fresh artifact namespace. The old ZIP with SHA-256 BF79073D...3FF640 is not reused.

2. THE ANNULAR 5:3 SCALAR
Let

$$
\begin{aligned}
R_X &= 5\, c_X(2) + 3\, c_X(3),\\
A_X &= R_X - R_{(X/4)},\\
H_Y(n) &= \min(\log 4, \log(Y/n))_+.
\end{aligned}
$$

The exact sparse scalar coefficients are

$$
a_*(n) = 6\, 1_{(n=1)} - 6\, \mu(n) + 9\, 1_{(2|n)}\, \mu(n/2) - 3\, 1_{(4|n)}\, \mu(n/4).
$$

Therefore

$$
A_X = \sum_{n\ge 1} a_*(n)\, H_X(n) / \sqrt{n}.
$$

Define

$$
\kappa_X(n) = -6\, H_X(n) + (9/\sqrt{2})\, H_{(X/2)}(n) - (3/2)\, H_{(X/4)}(n).
$$

Then the exact source identity is

$$
\begin{aligned}
A_X &= 6\, H_X(1) + C_X,\\
C_X &= \sum_{n\ge 1} \mu(n)\, \kappa_X(n) / \sqrt{n}. \qquad (1)
\end{aligned}
$$

The factors $9/\sqrt{2}$ and $3/2$ are forced by the substitutions $n=2m$ and $n=4m$. All sums are finite because the hinges vanish outside their endpoints.

3. THE CRITICAL FOUR-BAND BOUNDARY
Put

$$
B_{(1/2)}(t) = \sum_{n\le t} \mu(n)/\sqrt{n}.
$$

Finite Fubini gives, for every $Y>0$,

$$
\sum_{n\ge 1} \mu(n)\, H_Y(n)/\sqrt{n} = \int_{Y/4}^{Y} B_{(1/2)}(t)\, dt/t. \qquad (2)
$$

Substituting (2) into (1) and splitting the overlapping scale intervals yields

$$
\begin{aligned}
C_X = &-(3/2) \int_{X/16}^{X/8} B_{(1/2)}(t)\, dt/t\\
&+((9\sqrt{2}-3)/2) \int_{X/8}^{X/4} B_{(1/2)}(t)\, dt/t\\
&+((9\sqrt{2}-12)/2) \int_{X/4}^{X/2} B_{(1/2)}(t)\, dt/t\\
&-6 \int_{X/2}^{X} B_{(1/2)}(t)\, dt/t. \qquad (3)
\end{aligned}
$$

This is an exact identity for every real $X>0$; no asymptotic estimate enters.

Define C4MBI67(X), the critical four-band Mobius boundary inequality, by

$$
\text{C4MBI67}(X) \iff A_X \ge 0. \qquad (5)
$$

For $X\ge 4$ the threshold in (4) is simply $-6 \log 4$.

4. BLPTE67 IS THE SAME ROOT SIGN
Retain the exact T-97700 decomposition

$$U_{full}(X) = U_Z(X) - I_Z(X) - T_Z(X), \qquad (6)$$

where U_Z is the complete small-prime cube, I_Z is the closed terminal Type-I range, and T_Z is the signed large-prime Type-II term. At the root,

$$U_{full}(X) = A_X / \sqrt{X}. \qquad (7)$$

BLPTE67(X) is the proposed inequality

$$T_Z(X) \le U_Z(X) - I_Z(X). \qquad (8)$$

Combining (5)-(8) proves the exact root equivalence

$$\begin{aligned} &\mathrm{BLPTE67}(X) \\ &\iff U_{full}(X) \ge 0 \\ &\iff A_X \ge 0 \\ &\iff \mathrm{C4MBI67}(X). \end{aligned} \qquad (9)$$

Thus BLPTE67 is not a source-blind auxiliary estimate weaker than the producer. At root scope it is the producer sign in Bellman coordinates. The stronger state-wise formulation has additional quantifiers and is not silently claimed.

5. EXACT LOGARITHMIC PRIME OWNERSHIP

For every squarefree $n>1$,

$$\mu(n) \log n = - \sum_{p|n} \mu(n/p) \log p. \qquad (10)$$

Multiplying (10) by $\kappa_X(n)/(\sqrt{n} \log n)$, summing, and writing $n=pm$ gives

$$\begin{aligned} &\sum_{n>1} \mu(n)\, \kappa_X(n)/\sqrt{n} \\ &= - \sum_{p\le X} (\log p)/\sqrt{p} \\ &\qquad \sum_{m\le X/p,\ p \nmid m} \\ &\qquad \mu(m)\, \kappa_X(pm)/(\sqrt{m} \log(pm)). \end{aligned} \qquad (11)$$

Each squarefree product n is spent exactly once across its prime owners through the nonnegative weights

$$\log p / \log n, \quad p|n,$$

whose sum is one. There is no duplicated factorization and no unrestricted rough reservoir.

If B_X denotes the double sum on the right of (11) without its leading minus, then

$$A_X = 6 H_X(1) + \kappa_X(1) - B_X. \qquad (12)$$

For $X\ge16$,

$$\begin{aligned} &6 H_X(1) + \kappa_X(1) \\ &= ((9 \sqrt{2}-3)/2) \log 4. \end{aligned} \qquad (13)$$

Equations (11)-(13) give the exact signed prime-versus-Mobius formulation of the remaining one-sided inequality.

6. WHY UNSIGNED TYPE-II INFORMATION IS INSUFFICIENT

Take two atoms in the same declared block and owner class, with equal positive coefficient magnitudes and distinct kernel responses $K1>K2>0$. The sign vectors $(+1,-1)$ and $(-1,+1)$ have identical support, l1 norm, l2 norm, unsigned owner mass, and every source-blind large-sieve input. Their one-sided functionals are $K1-K2>0$ and $-K1+K2<0$. Therefore those unsigned data do not determine the sign of a nonconstant Type-II kernel.

The four-band kernel $\kappa_X(pm)/\log(pm)$ is nonconstant because its activation formula changes across the four bands. A valid proof of C4MBI67 must preserve an arithmetic covariance between the actual Mobius signs and that kernel. Taking rowwise absolute values before the pairing destroys the required information. This countermodel rejects a proof mechanism; it does not refute C4MBI67 itself.

7. ANALYTIC CONSUMER AND EXACT BOUNDARY

The inherited Mellin transform is

$$\begin{aligned} &\int_1^\infty A_X X^{-s-1}\, dX \\ &= (1-4^{-s}) \left[6/s^2 \right. \\ &\qquad \left. - 3(1-2^{-z})(2-2^{-z})/(s^2 \zeta(z)) \right], \\ &\quad z = s + 1/2. \end{aligned}$$

The finite numerator is zero-free in $\mathrm{Re}\, z>0$. Hence eventual positivity of A_X would imply RH by Landau's theorem. The present packet proves the exact form and equivalences of the remaining producer, not its sign.

THEOREM LEDGER

Four-band identity	PROVED EXACT
Logarithmic prime ownership	PROVED EXACT
BLPTE67(X) <=> C4MBI67(X) at root	PROVED EXACT
Source-blind unsigned Type-II implication	REFUTED
C4MBI67 sign	OPEN / RH-BEARING
Riemann Hypothesis	UNPROVED

REPLAY

The checker reconstructs the sparse coefficients on 500 controls, verifies all four band coefficients, checks logarithmic ownership on 305 squarefree cases, checks 125 Bellman-equivalence fixtures, and runs eight mutation/status tests. Expected verdict:

PASS_T97701_BLPTE_C4MBI67_CRITICAL_CORE_ALGEBRA

Proof-object SHA-256:

8c65f8e1a340510c153688196de0782938eb377277cc425808b53e5d931b1f6d